

NIZAMUDHEEN T I

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SUMMARY

Experienced Data Scientist specializing in Generative AI, Synthetic Data, and Knowledge Graph-driven analytics. Skilled in designing, fine-tuning, and deploying LLM-based systems with RAG pipelines and agentic AI workflows. Strong expertise in data science practices, ML model development, and LLM-based applications, backed by a Master's in Artificial Intelligence with top academic distinction.

WORK EXPERIENCE

Data Scientist | Prevalent AI, Cochin

July 2023 – Present

- Designed and implemented client-facing security dashboards to compare global vulnerability trends with internal findings, identify emerging high-risk patterns and baseline vulnerabilities across shared technologies, and correlate vulnerability reduction with technology stacks to evaluate remediation effectiveness and enable data-driven risk prioritization.
- Developed an LLM-based system for compensating control mapping in vulnerability management, enabling security teams to automatically identify relevant controls and quantify reductions in security risk.
- Executed a Proof-of-Concept for Agentic AI workflows to establish an Agentic Data Fabric, enhancing data source onboarding and security insight delivery. Accelerated internal and client development workflows using remote AI agents and Model Context Protocol (MCP) servers.
- Led data science and LLMOps efforts for a GenAI-powered mobile and web security platform, leveraging advanced RAG methods and context engineering to deliver real-time risk insights. Implemented robust monitoring and AI guardrails to ensure Responsible AI and secure data governance.
- Fine-tuned Small Language Models for domain-specific use cases, including software extraction, asset role classification, and vulnerability consequence prediction.
- Optimized SLM inference for production GenAI workflows using vLLM with KV and prompt caching, along with PEFT (LoRA) and model quantization, significantly reducing latency, token recomputation, memory usage, and compute cost while preserving model accuracy.
- Designed and implemented a Data Quality measurement framework to assess multiple Data Quality dimensions, helping users and developers evaluate data quality and its impact on delivering reliable insights to stakeholders.

- Designed and implemented an agent-based synthetic data generation module to create highly correlated multidimensional data across cybersecurity sources, enabling faster product development and performance testing.
- Built a text feature extraction system to parse temporal expressions in chat data, improving NLP pipeline accuracy for downstream chatbot applications.
- Applied topic modeling and anomaly detection on SQL query logs to detect injection attempts, strengthening system security and reliability.

EDUCATION

- **Master of Technology in Artificial Intelligence** [2020-22]
College of Engineering, Trivandrum – CGPA: 9.64
- **Advanced Diploma in Artificial Intelligence** [2018-19]
National Institute of Electronics and Information Technology, Calicut – CGPA: 9.2
- **Bachelor of Technology in Computer Science & Engineering** [2014-18]
LBS College of Engineering, Kasaragod – CGPA: 6.78

TECHNICAL SKILLS

- **Programming:** Python, R, SQL
- **Generative AI & LLMOps:** LangChain, LangGraph, MCP, A2A, LangFuse, AWS Bedrock, Azure OpenAI, Transformer, PEFT
- **Data Science & ML:** Scikit-learn, TensorFlow, PyTorch, Keras, MLFlow
- **Data Engineering & Visualization:** Pandas, PySpark, NumPy, Matplotlib, Seaborn
- **Cloud & Deployment:** AWS, Azure, FastAPI, Docker, ArgoCD, CI/CD Pipelines, Streamlit
- **Databases/Vector Databases:** PostgreSQL, Redis, Milvus, FAISS
- **Version Control & DevOps:** Git, GitHub Actions

CERTIFICATIONS

- Architecting AI Solutions (Coursera)
- Advanced LLMOps (LinkedIn)
- Generative AI with LLMs (DeepLearning.ai)
- IBM AI Engineering & Data Science (Coursera)

ADDITIONAL INFORMATION

Languages: English, Malayalam